Predicting Diabetes in Pima Indian Women

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Business Problem

Diabetes is a chronic disease that imposes significant health and economic burdens on individuals and healthcare systems. Complications such as cardiovascular disease, kidney failure, and vision loss can often be reduced through early detection and intervention. However, screening all patients with comprehensive clinical tests is costly and time consuming. This project addresses the problem of using routinely collected clinical data to predict diabetes risk and support evidence-based screening decisions, enabling clinicians to prioritize high-risk individuals for further testing and preventive care.

Background / History

The prevalence of diabetes has increased worldwide over recent decades, driven by demographic changes, urbanization, reduced physical activity, and rising obesity rates. The Pima Indian population has been extensively studied in the medical literature because of historically high rates of type 2 diabetes. As a result, the Pima Indians Diabetes Dataset has become a benchmark dataset in machine learning research for evaluating classification methods in healthcare. Using this dataset allows comparison with prior work while also demonstrating how modern data science methods can support clinical decision-making.

Data Explanation (Data Preparation / Data Dictionary)

The analysis uses the Pima Indians Diabetes Dataset, which contains eight clinical and demographic predictors: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age. The binary outcome variable indicates diabetes status. Several variables contain invalid zero values that are not physiologically plausible. These values were treated as missing and replaced using median imputation. Outliers were examined using distribution plots, and features were standardized for models sensitive to scale. A detailed data dictionary describing each variable is provided in Appendix A.

Methods

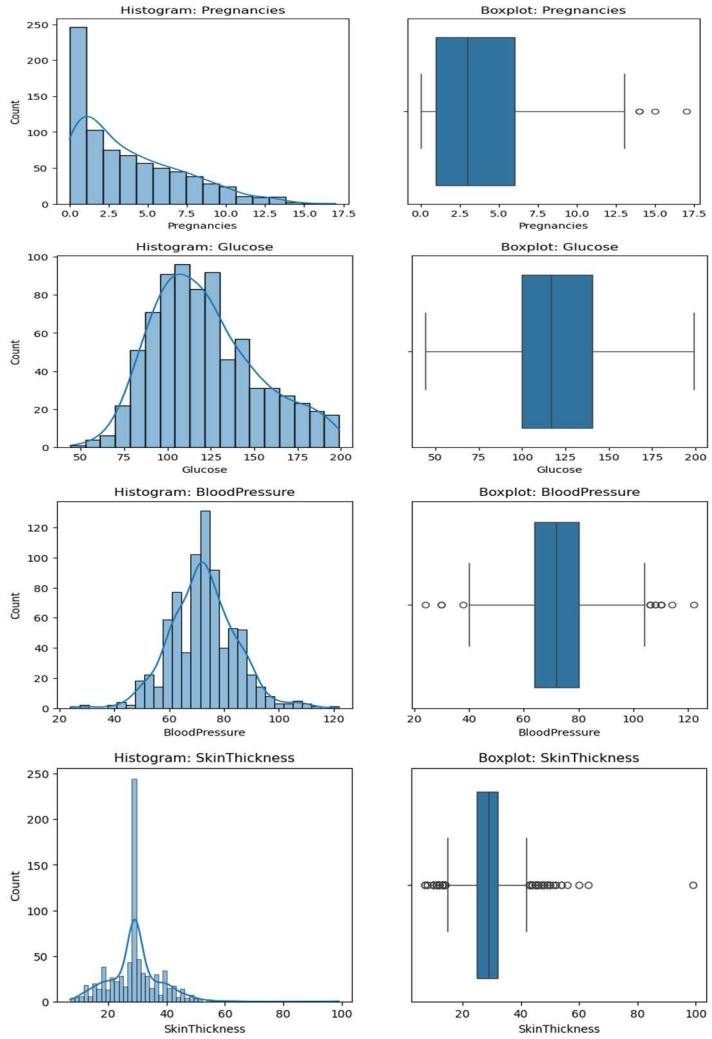
Exploratory data analysis (EDA) was conducted using summary statistics and visualizations to understand distributions, detect skewness, and identify potential predictors. Correlation analysis was used to assess linear relationships and possible redundancy among features. Two supervised learning models were implemented: Logistic Regression as a baseline interpretable model and Random Forest as a nonlinear ensemble method. Models were evaluated on a hold-out test set using accuracy, precision, recall, F1-score, and the Area Under the ROC Curve (AUC). In addition, confusion matrices were used to assess the types of classification errors.

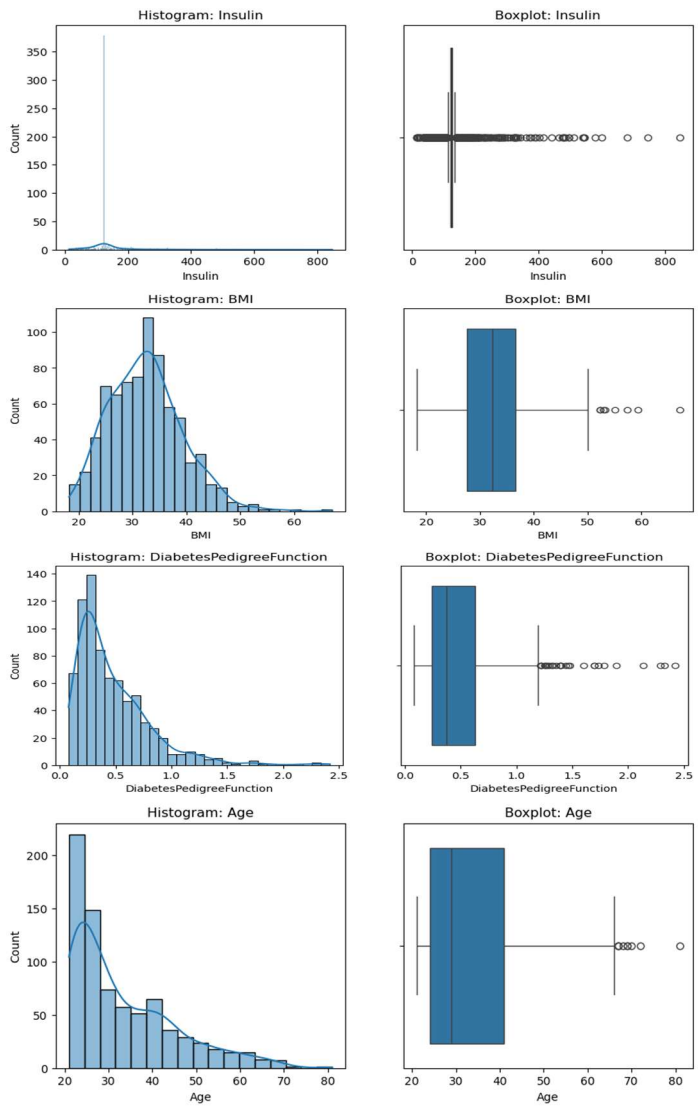
Analysis

The EDA results reveal that several variables are right-skewed, particularly Insulin and Skin Thickness, indicating the presence of extreme values and reinforcing the need for robust preprocessing. Histograms show the overall distributions of predictors and highlight class imbalance in the outcome variable. The boxplot of Glucose by Outcome demonstrates a clear separation between groups, with diabetic patients showing substantially higher median glucose levels. The correlation heatmap shows that Glucose has the strongest association with the diabetes outcome, followed by BMI, Age, and Pregnancies. Model evaluation further supports these findings. The Logistic Regression model achieves approximately 75% accuracy with an AUC around 0.82, indicating good discriminative performance. The confusion matrix shows stronger performance for non-diabetic cases and moderate sensitivity for diabetic cases, highlighting the importance of improving recall. Random Forest feature importance ranks Glucose as the most influential predictor, followed by BMI, Age, and Diabetes Pedigree Function, aligning with medical understanding and earlier exploratory analysis.

Figure Interpretation and Discussion

The visualizations support the modeling results. The histograms show the distribution of the variables and indicate skewness in features such as Insulin and Skin Thickness.
The boxplot comparing glucose by diabetes outcome shows that individuals with diabetes generally have higher glucose levels.





The correlation heatmap highlights Glucose, BMI, and Age as important predictors.

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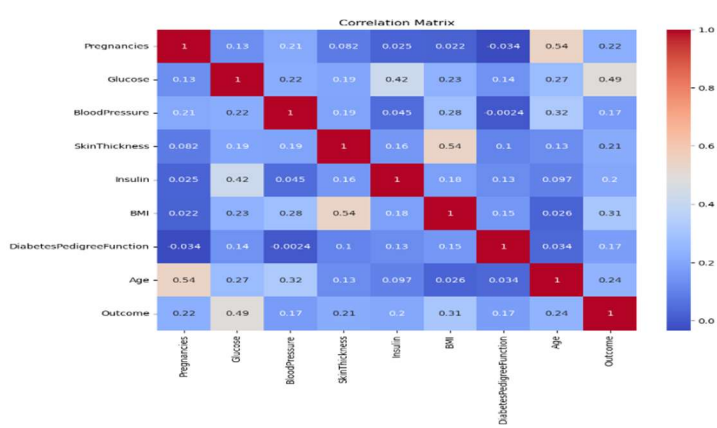
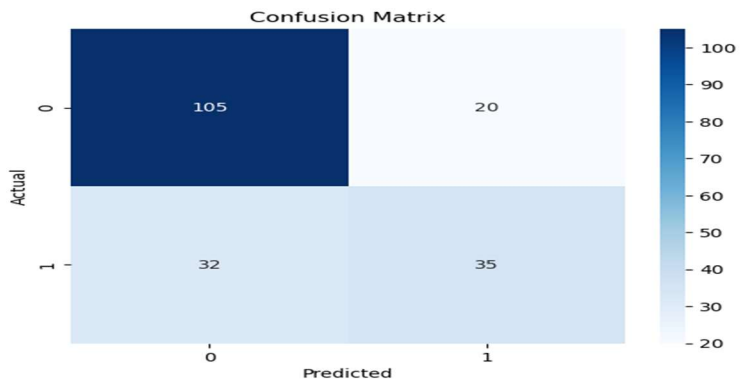
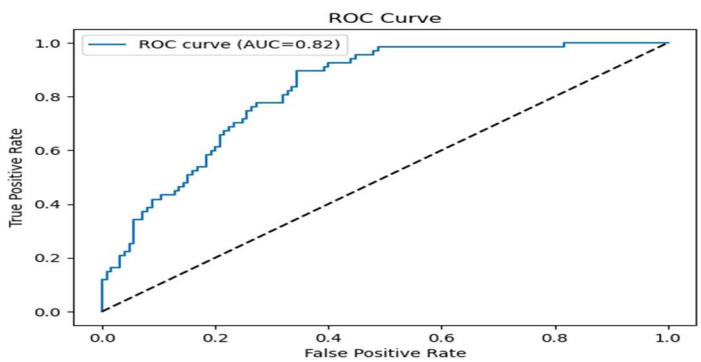


The confusion matrix shows that the model correctly classifies most non-diabetic cases but still produces some false negatives.



Finally, the feature importance plot and ROC curve confirm that Glucose is the strongest predictor and that the model performs well with an AUC of about 0.82.



Conclusion

This study demonstrates that machine learning models can effectively predict diabetes risk using commonly available clinical measures. The combination of EDA, correlation analysis, and supervised modeling provides consistent evidence that Glucose and BMI are the most informative predictors. While performance is not perfect, the results suggest that such models can serve as useful screening support tools when combined with clinical judgment.

Assumptions

The analysis assumes that clinical measurements are accurate and consistently collected, that the dataset is representative of the studied population, and that relationships observed in the data remain stable over time. It also assumes that median imputation does not introduce systematic bias and that the selected features capture the most relevant risk factors for diabetes.

Limitations

The dataset contains a limited number of features and omits important lifestyle variables such as diet and physical activity. Class imbalance may bias performance metrics toward the majority class. Findings derived from this specific population may not generalize to other demographic groups without further validation.

Challenges

Key challenges include handling missing and noisy data, improving sensitivity for diabetic cases, balancing interpretability with predictive performance, and communicating uncertainty and limitations to non-technical stakeholders in a healthcare context.

Future Uses / Additional Applications

Future work could integrate electronic health records, longitudinal patient data, lifestyle information, or genetic markers to improve predictive performance. The same analytical framework could also be applied to other chronic disease risk prediction tasks.

Recommendations

Healthcare providers should prioritize glucose and BMI in routine screening protocols. Before deployment, models should be calibrated to emphasize recall for diabetic cases. Additional data collection and external validation on broader populations are recommended to improve robustness and generalizability.

Implementation Plan

Implementation would involve integrating the model into clinical decision support systems, training healthcare staff to interpret outputs, piloting the system in a controlled clinical environment, monitoring real-world performance, and establishing regular retraining schedules to account for data drift.

Ethical Assessment

Ethical considerations include protecting patient privacy, ensuring data security, mitigating bias against specific groups, and carefully managing the consequences of misclassification. Predictive tools should not support clinical judgment, and governance frameworks should ensure transparency and accountability.

Appendix A: Data Dictionary

Pregnancies: Number of times pregnant.

Glucose: Plasma glucose concentration.

BloodPressure: Diastolic blood pressure (mm Hg).

SkinThickness: Triceps skin fold thickness (mm).

Insulin: 2-Hour serum insulin (mu U/ml).

BMI: Body mass index (kg/m^2).

DiabetesPedigreeFunction: Genetic risk score proxy.

Age: Age in years.

Outcome: 1 = Diabetes, 0 = No diabetes.

Responses to Audience Questions

1. How reliable is this model in real clinical settings?

The model shows good discriminative performance with an AUC of approximately 0.82 and an accuracy around 75%, which suggests it can be useful as a screening support tool. However, real-world reliability requires external validation on new populations, prospective testing, and continuous monitoring.

2. What are the consequences of false negatives and false positives?

False negatives may delay diagnosis and treatment, while false positives may lead to unnecessary follow-up tests and anxiety. In healthcare, false negatives are typically more concerning, so improving recall is a priority.

3. Why were these models selected over alternatives?

Logistic Regression was chosen for interpretability and clinical familiarity, while Random Forest was chosen for its ability to model nonlinear relationships and provide feature importance, offering a balance between transparency and performance.

4. How does class imbalance affect performance?

Class imbalance can bias models toward predicting the majority class, making accuracy misleading. This is why recall, precision, and AUC were also used to evaluate performance.

5. Can the model generalize to other populations?

Because the dataset represents a specific population, generalization is not guaranteed. External validation on diverse populations is necessary before broader clinical use.

6. What additional data would most improve accuracy?

Lifestyle factors such as diet, physical activity, and smoking status, as well as longitudinal clinical data, would likely improve predictive performance.

7. How often should the model be retrained?

The model should be retrained periodically as new data become available, for example annually or whenever performance monitoring indicates degradation.

8. How will patient privacy be protected?

Data should be de-identified, securely stored, and accessed only by authorized personnel, in compliance with relevant data protection regulations and institutional policies.

9. How would this integrate into existing clinical workflows?

The model could be embedded in electronic health record systems as a decision support tool that flags high-risk patients for clinician review.

10. What validation is required before deployment?

Before deployment, the model should undergo external validation, prospective clinical evaluation, and bias and fairness assessments, along with appropriate regulatory and institutional review.

References

UCI Machine Learning Repository. (n.d.). Pima Indians Diabetes Database. Kaggle. (n.d.). Pima Indians Diabetes Dataset. Additional academic sources on diabetes prediction and machine learning methods will be used to support the analysis.